Math 9/10 - Factors and Polynomials Practice Test

Rubric								
Outcome	0	1	2	3	4	5	6	7
Understanding and Solving								
Mathematical Communication								

St. George's School - Academic Integrity

Academic Dishonesty includes any conduct with the intent to gain an unfair advantage in connection with an academic assessment. Academic Dishonesty can occur in many ways. Common forms of Academic Dishonesty include, but are not limited to, cheating, falsification, plagiarism, and tampering. Cheating occurs when an individual undermines the integrity of an assessment (including homework and other assignments, reports, projects, quizzes, tests, exams, or other forms of performance evaluations).

Examples of cheating include:

- Copying any task of an assessment;
- Allowing others to copy any task of an assessment;
- Improperly giving or receiving assessment information;
- Using unauthorized resources for or during an assessment;
- Skipping classes to avoid an assessment

Name : _____

Teacher's name: _____

Formulas

$a_n = a + (n-1)d$ $\qquad$ $S_n = \frac{n}{2}(a+l)$ $\qquad$ $S_n = \frac{n}{2}[2a + (n-1)d]$

Level 1 to Level 2 Questions

1. Find the GCF and LCM of the two numbers 100 and 125. Show how you got to the answer.

2. Given the prime factorization of the below three numbers, find

$$54 = 2 \times 3^3$$
$$72 = 2^3 \times 3^2$$
$$150 = 2 \times 3 \times 5^2$$

a) $GCF(54, 72, 150)$ b) $LCM(54, 72, 150)$

3. Complete the table below for each polynomial.

Expression	**Classification**	**Leading Coefficient**	**Degree**
$20xy + x^5y - 6$			
$2xy^2 + xy$			
45			

4. Simplify $6e + 4g + e^2 - 10e + 4e^2 - 9g$

5. Add or subtract the following polynomials.

a) $(3x^2 - 4x - 5) + (6x^2 - 8x)$

b) $(5x - 2x^2) - (2x^2 - 7)$

6. Multiply the following polynomials

a) $-3x \cdot 5x^4$

b) $-3xy(4x^2y + 3xy^2 - y)$

7. Divide the following polynomials.

a) $\dfrac{24x^2 - 12x + 6}{6}$

b) $\dfrac{12x^3 + 8x^2 - 24x}{4x}$

Level 3 to Level 4 Questions

8. Expand and simplify the following.

   a) $(x-7)(x+5)$

   b) $(3x-1)(2x^2-2x-3)$

9. Perform the combined operations and simplify.

   a) $\dfrac{-4x^2(2+x)}{2x} + \dfrac{3x(6x-8)}{2}$

   b) $-2(x+3) - (x+4)(x-4)$

10. Factor $3x^2 - 9x - 2xy + 6y$ by grouping.

11. Fully factor the following, if possible.

a) $x^2 - 12x - 28$

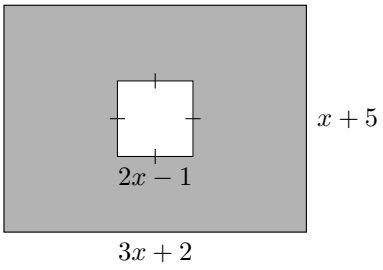
b) $x^2 + 10x + 16$

c) $16x^2 - 81y^2$

d) $44 - 18x - 2x^2$

Level 5 to Level 6 Questions

12. Find a simplified expression in terms of x to represent the shaded area.



Ans: _____

13. Fully factor the following, if possible.

a) $6x^2 + 5x - 4$

b) $10x^2 - 13x + 4$

c) $16x^4 - y^4$

d) $-15x^3 + 55x^2 + 20x$

14. For each of the following, find the value(s) of b that

a) Makes $3x^2 + bx - 10$ factorable.

b) Makes $25x^2 + bx + 49$ a perfect square trinomial.

Level 7 Questions

15. Answer the following parts.

    a) **Warm-Up:** What is the degree and all the coefficients of the polynomial $x^2 + 2x + 3$?

    b) **Warm-Up:** Let $f(x) = 3x^3 + 4x^2 - 7x + 6$, find $f(1)$.

    c) We define the polynomial function, $P(x)$, as an n-th degree polynomial written as

    $$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$$

    where $a_n, a_{n-1}, \ldots, a_0$ are the coefficients of the polynomials, and $a_n \neq 0$.

    If the coefficients $a_n, a_{n-1}, \ldots, a_0$ form an arithmetic sequence with $a_n = 100$ and $d = -3$.

    If $P(x)$ is a 30th degree polynomial (x^{30} is the highest power).

    Find $P(1)$ using the arithmetic sequence and series formulas.